FinInsight

FINANCIAL ANALYTICS PROJECT

Expense Control and Profitability Analytics

Data Analytics Internship
Project Report

Submitted to:

Persevex

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TOOLS & TECHNOLOGIES

XL	PQ	SQL	BI	DAX
Excel	Power Query	PostgreSQL	Power BI	DAX

DATASET

Microsoft Contoso Financial Sample Data

OCTOBER 2026



01 | Executive Summary

Financial performance, reporting approach, and principal findings

PROJECT OVERVIEW

FinInsight - Expense Control and Profitability Analytics analyzes expense balances, budget allocation, departmental and regional performance, and forecast scenarios. The project combines interactive Power BI dashboards with independent SQL-based financial analysis.

Data and analytical approach

The historical Microsoft Contoso financial sample contains 541,522 fact records, supported by 641 account and 2,036 geography dimension records. Excel and Power Query supported preparation; PostgreSQL and SQL supported validation; Power BI and DAX supported interactive reporting.

Dashboard and analysis

The six-page Power BI report includes a landing page, executive overview, expense and budget analysis, departmental and regional analysis, profitability-related account review, and forecast trends. Results were reconciled with SQL calculations using matching reporting filters.

Key findings

For July-December 2012, filtered to YTD expense records in the N/A scenario, actual expense balances totaled $626.72M against a budget of $744.93M. Payroll and Outsourced Resources were the largest expense contributors; Other People and Product Development displayed positive budget variances.

Key Financial Highlights

ACTUAL EXPENSES	EXPENSE BUDGET
$626.72M	$744.93M
EXPENSE VARIANCE	**BUDGET UTILIZATION**
-$118.21M	84.13%

INTERPRETATION

Expense variance represents signed actual less signed budget; this is not automatically realized cash savings. Forecast scenario differences are not treated as verified forecast accuracy.

02 | Project Objectives

Practical financial questions translated into an analytical solution

PRIMARY OBJECTIVE

Build a consistent, interactive financial analytics solution that supports expense monitoring, budget evaluation, and transparent decision-making through validated SQL measures and Power BI reporting.

Specific objective

01	**Expense Monitoring** Track signed actual expense balances and identified major spending contributors.
02	**Budget Evaluation** Measure actual-versus-budget differences and monitor budget utilization.
03	**Department and Region** Compare expenses across functional areas, profit centers, and regions.
04	**Monthly Trends** Explore changes in financial balances across the reporting period.
05	**Forecast Comparison** Examine actual and October Forecast scenarios and qualify coverage gaps.
06	**Interactive Power BI** Create navigable KPI views, filters, charts, and summary matrices.
07	**SQL Verification** Validate joins, key uniqueness, totals, and analytical outputs in PostgreSQL.

SUCCESS CRITERION

Key Power BI expense figures should reconcile with independent PostgreSQL queries under equivalent date, reporting-status, and account-class filters.

03 | Dataset Description

Microsoft Contoso historical financial sample

The model follows a financial fact plus two dimensions structure. Dimensions provide descriptive attributes used to slice measures by account category, function, and geography.

Three related data tables

TABLE	RECORDS	DESCRIPTION
FactFinanceData	541,522	Monthly Actual, Budget, Forecast, reporting flags, Account and Profit Center
DimAccounts	641	Account hierarchies: Class, Group, Sub Class, and Line Item
DimExecGeography	2,036	Profit Center, functional area, organization, and worldwide region

Relational model

ACCOUNT DIMENSION	FINANCIAL FACT	GEOGRAPHY DIMENSION
DimAccounts	FactFinanceData	DimExecGeography
Join key: Account	Account + Profit Center	Join key: Profit Center
One matching dimension row	Many finance records	One matching dimension row

Fields used in analysis

ROLE	FIELDS	PURPOSE
Time	Fiscal Month / FiscalDate	Monthly reporting and trends
Identifiers	Account; Profit Center	Links to dimension attributes
Scenarios	Forecast Version; YTD Flag	Distinguish actual, budget and forecast scenarios
Measures	Actual; Budget; Forecast	Calculate financial KPIs
Categories	Class; Group; Sub Class	Expense category and composition
Geography	WW Region; SEC Func Area	Regional and departmental performance

DATASET PROVENANCE

Microsoft Contoso financial sample, not Persevex financial records. All dollar figures describe historical sample balances.

04 | Tools & Technologies

An end-to-end analytics workflow

01	**Excel - Source inspection** Reviewed table structure and prepared UTF-8 CSV exports.
02	**Power Query - Data preparation** Inspected column profiles, data types, and transformation steps.
03	**PostgreSQL - Analytical database** Stored the three related tables and supported reconciliation.
04	**SQL - Financial queries** Validated joins and calculated expenses, variances, and monthly trends.
05	**Power BI + DAX - Reporting** Built six pages, dynamic financial KPIs, and navigation.

Saved SQL deliverables

SCRIPT	PURPOSE
01_create_tables.sql	Table definitions and column data types
02_data_validation.sql	Row counts, joins, and duplicate-key checks
03_expense_budget_analysis.sql	Actual-versus-budget comparison
04_monthly_trends.sql	Monthly expenses, budget and variance
05_department_region_analysis.sql	Functions and worldwide regions
06_forecast_analysis.sql	Monthly scenarios and record-coverage checks
07_business_insights.sql	Category drivers and positive budget variance

REPRODUCIBILITY

SQL queries were stored as named .sql files, and core Power BI measures were reconciled with PostgreSQL under equivalent filters.

05 | Data Cleaning & Quality

Profile, import, validate, and reconcile before drawing conclusions

Preparation workflow

01 - SOURCE	02 - PROFILE	03 - EXPORT	04 - IMPORT	05 - VERIFY
Contoso Access tables	Power Query checks	UTF-8 CSV files	PostgreSQL tables	SQL joins and KPIs

Quality controls applied

01	**Column mapping** Verified original CSV header order before creating SQL tables.
02	**Key uniqueness** Checked distinct Account and Profit Center identifiers.
03	**Duplicate imports** Rebuilt the PostgreSQL environment after an initial duplicate-key issue.
04	**Relationship tests** Used LEFT JOIN checks for unmatched Account and Profit Center keys.
05	**Source reconciliation** Matched July-December 2012 expense KPIs with Power BI.
06	**Scenario coverage** Compared actual and forecast record availability before interpretation.

IMPORT CORRECTION DURING DEVELOPMENT

An initial SQL import placed Account hierarchy values in the wrong columns because the CSV header order differed from the table definition. The database was recreated using Account, Line Item, Group, Sub Class, Class.

DATA-HANDLING RULE

Financial fact rows may share Account and Profit Center while differing by month, version, or YTD flag. Do not deduplicate those keys alone.

06 | Power BI Dashboard

Six connected pages with consistent FinInsight design

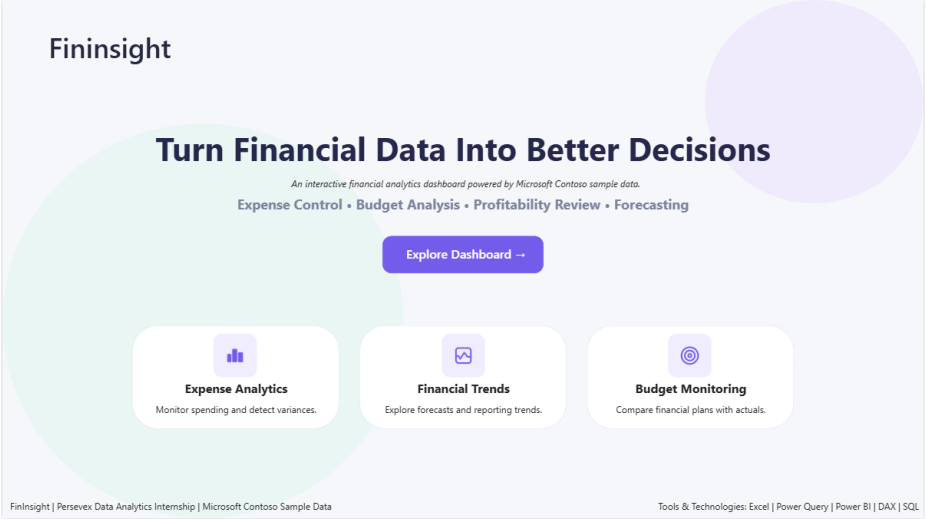


Figure 1: Home Page

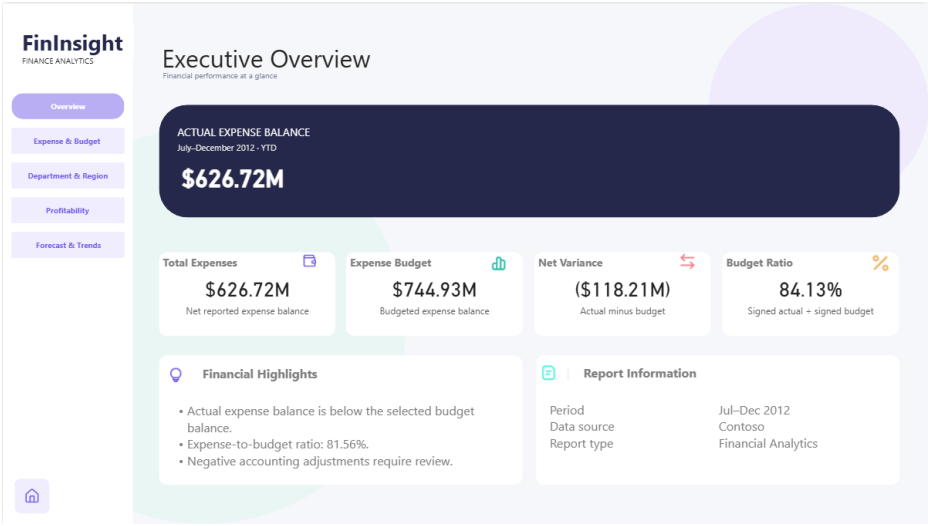


Figure 2: Overview Page

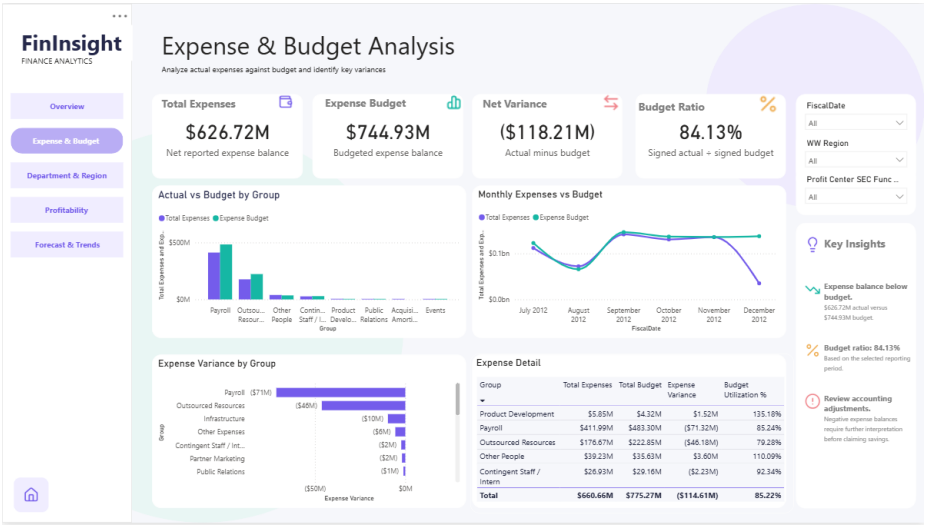


Figure 3: Expense & Budget Page

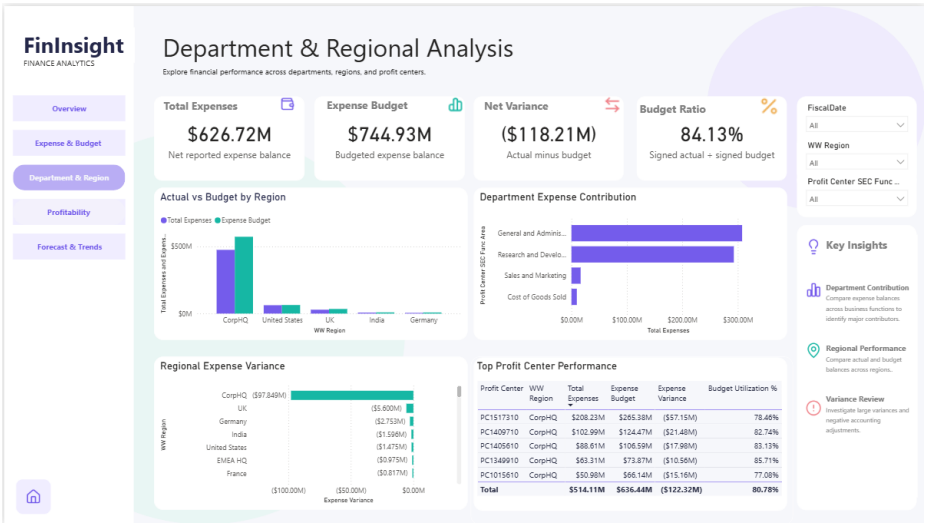


Figure 4: Department & Region Page

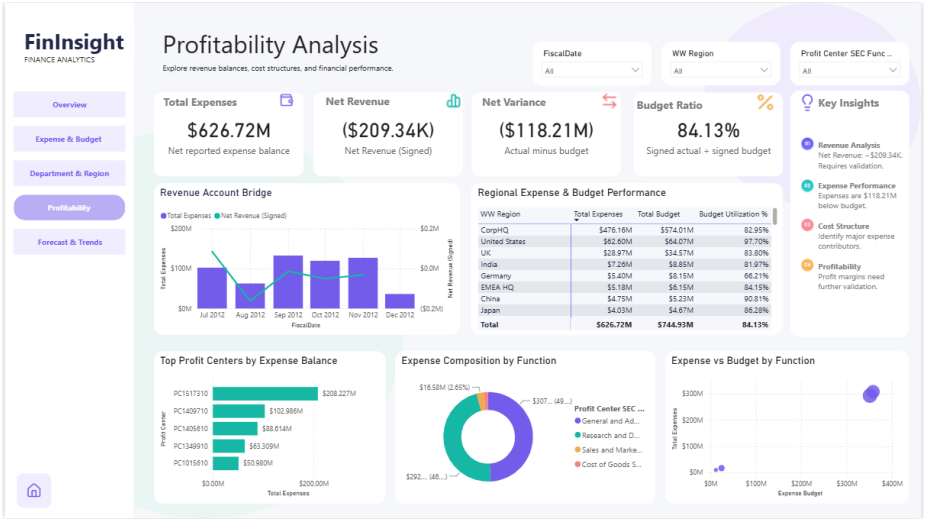


Figure 5: Profitability Page

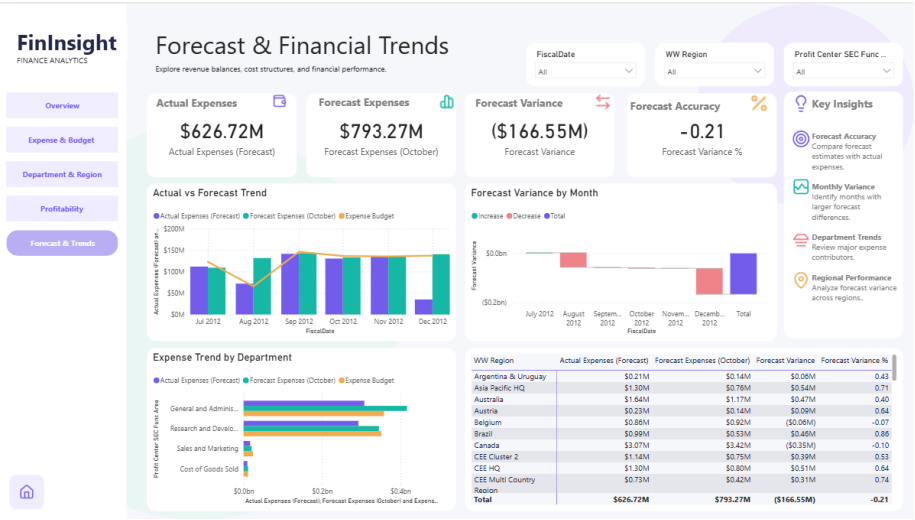


Figure 6: Forecast & Trends Page

Report navigation

PAGE	REPORT AREA	ANALYTICAL ROLE
01	Home	Landing page and navigation
02	Overview	Expense and budget KPIs
03	Expense & Budget	Variance and monthly trends
04	Department & Region	Functional and regional comparisons
05	Profitability	Signed balances and cost composition
06	Forecast & Trends	Monthly scenario differences

07 | SQL Analysis & Validation

Independent verification of financial reporting calculations

Validated data model

TABLE / TEST	RECORDS	PURPOSE
FactFinanceData	541,522	Main financial fact records
DimAccounts	641	Distinct account dimension rows
DimExecGeography	2,036	Distinct profit-center rows
Join result	541,522	No multiplication after corrected import

Reporting filters reproduced in SQL

SELECTED REPORTING SLICE

Class: Expenses | Forecast Version: N/A | YTD Flag: YTD
Period: July-December 2012 | Matching expense records: 130,250

Representative PostgreSQL calculation

```
SELECT
  ROUND(SUM(f."Actual") / 1000000, 2) AS actual_m,
  ROUND(SUM(f."Budget") / 1000000, 2) AS budget_m,
  ROUND((SUM(f."Actual") - SUM(f."Budget")) / 1000000, 2)
    AS variance_m
FROM public.factfinancedata AS f
JOIN public.dimaccounts AS a
  ON f."Account" = a."Account"
WHERE a."Class" = 'Expenses'
  AND f."Forecast Version" = 'N/A'
  AND f."YTD Flag" = 'YTD'
  AND to_date(f."Fiscal Month", 'FMMonth, YYYY')
    BETWEEN DATE '2012-07-01' AND DATE '2012-12-01';
```

Cross-tool reconciliation

ACTUAL EXPENSES	EXPENSE BUDGET
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EXPENSE VARIANCE	**BUDGET UTILIZATION**
-$118.21M	84.13%

08 | Financial Findings

July-December 2012 | YTD | Expenses | Forecast Version: N/A

ACTUAL EXPENSES	BUDGET	VARIANCE	UTILIZATION
$626.72M	$744.93M	-$118.21M	84.13%

Largest expense contributors

EXPENSE CATEGORY	ACTUAL $M	BUDGET $M	VARIANCE $M
Payroll	411.99	483.30	-71.32
Outsourced Resources	176.67	222.85	-46.18
Other People	39.23	35.63	+3.60
Product Development	5.85	4.32	+1.52

Monthly Actual and October Forecast balances

MONTH	ACTUAL $M	FORECAST $M	VARIANCE %
July 2012	111.86	109.19	+2.44%
August 2012	72.08	131.71	-45.27%
September 2012	141.57	143.30	-1.21%
October 2012	130.61	133.24	-1.97%
November 2012	135.56	135.16	+0.29%
December 2012	35.05	140.66	-75.08%

FORECAST INTERPRETATION

The largest monthly scenario differences were August (-45.27%) and December (-75.08%). These differences are scenario variances, not confirmed forecast accuracy errors; record coverage differs between scenarios.

09 | Business Insights

Budget deviations prioritized by monetary impact and analytical context

CORE FINDING

Aggregate actual expense balance was $118.21M below budget. This does not exclude positive category-level variances and should not automatically be interpreted as realized cash savings.

Categories with positive budget variances

CATEGORY	ACTUAL $M	BUDGET $M	VAR $M	VAR %
Other People	39.23	35.63	+3.60	+10.09%
Product Development	5.85	4.32	+1.52	+35.18%
Relationship Marketing	0.45	0.03	+0.42	+1338.71%
Localization	0.43	0.06	+0.38	+686.81%
Advertising	0.25	0.08	+0.18	+230.28%
Data Center Hosting	0.14	0.08	+0.06	+82.76%

Actionable recommendations

01	**Workforce and contractors** Review staffing, contractor use, and the drivers of large, signed differences.
02	**Positive category variances** Investigate Other People (+$3.60M) and Product Development (+$1.52M).
03	**Small-budget percentages** Consider monetary size alongside large marketing and localization percentages.
04	**Monthly budget controls** Document material month-on-month changes in actual-versus-budget balances.
05	**Forecast comparability** Align account coverage and observed periods before claiming forecast accuracy.

10 | Challenges & Limitations

What the analysis confirms - and what it does not

Challenges addressed during the project

01	**Large financial dataset** Processed a fact table with more than half a million rows and kept the reporting model efficient.
02	**CSV-to-database mapping** Corrected an initial Account column-order mismatch when rebuilding PostgreSQL tables.
03	**Duplicate keys during import** Identified and resolved a duplicate-dimension import that multiplied JOIN results.
04	**Consistent financial filters** Matched date, YTD, forecast version, and Expenses filters in PostgreSQL and Power BI.
05	**Signed financial balances** Avoided assuming negative allocation balances represented cash savings or profit.
06	**Different scenario coverage** Measured Actual-versus-Forecast key overlaps and its effect on interpretation.

Limits on interpretation

FORECAST

Only about 59%-62% of October-December forecast Account / Profit Center keys were present in the actual scenario. Scenario variance alone is not an accuracy score.

PROFITABILITY AND SCOPE

Signed net revenue and expense allocations do not establish a conventional, verified profit margin. Historical Contoso sample data demonstrates analytical methods rather than Persevex financial performance.

11 | Conclusion & Future Scope

The outcome of the FinInsight analytics project

PROJECT OUTCOME

FinInsight demonstrates how Excel, Power Query, PostgreSQL, SQL, Power BI, and DAX can be combined to transform a historical financial dataset into a navigable analytical report. The project reconciled key expense and budget KPIs across SQL and Power BI and surfaced category, regional, and scenario-level differences for review.

Skills and deliverables

01	**Data Preparation** Inspected source records, mappings, and reporting fields.
02	**SQL Analytics** Wrote repeatable joins, aggregations, comparisons, and validations.
03	**Business Intelligence** Built a six-page Power BI report with KPI measures and navigation.
04	**Financial Communication** Translated validated results into qualified findings and recommendations.

Future scope

- Automate source refresh and repeatable SQL-to-dashboard validation.
- Expand fiscal period, department, and allocation-level comparisons.
- Build true forecast-accuracy measures only with confirmed forecast timing and comparable actual records.
- Investigate revenue sign conventions and verified cost allocations before deriving profit margins.

PROJECT DETAILS

Prepared by Shouary Verma | October 2026 | Microsoft Contoso Financial Sample Data